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CAJ Theory Assignment

Q1 What is the difference between a database schema and a database state?

Database Schema

→ Database schema refers to the overall logical structure or design of a database, defining how data is stored/organised, what table exist, what attribute/columns each table has, their datatypes, relationships & constraints b/w tables

→ created during database design & does not change frequently.

→ Change occasionally when database structure is modified, relatively static in nature.

→ Defined by Database designers or administrators.

→ Ex- Student table with St_ID, Name, Age & Course table.

→ Also known as "Intension".

Database State

→ Database State is a database instance or snapshot, refers to actual data stored in db at a certain point of time

→ exists at any point of time between db creation & deletion

→ state changes frequently whenever records are inserted/updated / deleted. hence dynamic in nature

→ defined by users and applications through db operations.

→ Ex - records of student/course saved in respective tables.

→ Also known as "Extention".

→ Initial state = refers to db when it is loaded
→ Valid state = refers to state that satisfies structure & constraints of the database.

Q-2 What are various categories of Data Models?

→ Data Model
↳ collection of concepts used to describe → the structure of the DB
→ relationships b/w data
→ Constraints on data
→ operations performable on data.
↳ help DB designers represent real world info. in structured manner - before implementing in database system.
→ broadly classified into following categories

1. Conceptual / High level Data models
→ describe the DB at a high level. focusing on entities, their attributes and relationships between them.
→ mainly used during initial database design phase b.c are easily understandable for both technical & non technical users.
→ most common example Entity-Relationship (ER) Model
Q. An ER model has:
→ Entities : real world objects
→ Attributes : Properties of entities
→ Relationships : Association between entities.
purpose: to understand & represent the requirements of DB before deciding how the data will actually be stored.

2. Logical / Representational Data Models:
→ describes the DB structure & how data is logically organised in DB
represent records, attributes, relationships & constraints.
→ Major types:-
a) Relational Data Model:
→ Represents data in the form of tables (relations)
→ Rows represent records, while columns represent attributes
→ Primary keys uniquely identify records.
→ Foreign keys establish relationships between tables.
b) Hierarchical Data Model:
→ Organises data in tree-like structure
→ Each child record generally has only one parent.
→ A parent can have multiple child records.
→ Suitable for representing hierarchical relationships.
c) Network Data Model:
→ Represents Data as records connected through links.
→ A child record can have multiple parent records
→ Supports many-many relationships.
d) Object Oriented Data Model:
→ Represents data in the form of objects.
→ Objects contain both attributes and methods.
→ Supports object-oriented concepts such as inheritance & encapsulation.
→ Useful for complex applications involving multimedia & engineering data.

3. **Physical / Low level Data Model :**

→ describes how data is actually stored in computer storage.

→ include details such as:

- File Organization
- Record storage formats
- Indexing techniques
- Storage block and pages
- Access paths.

→ Data models provide different levels of abstraction for DB design. Conceptual models describe what data is required, Logical models describe how data is organised & physical models describe how data is stored.

Q-3 Differentiate File processing & database approach.

→

	File processing Approach	Database Approach
Data storage :	Data stored in seprate files	Data is stored in a structured database
Data Redundancy	High chance of duplicate data	redundancy can be reduced
Data Consistency	Difficult to maintain consistency across file	Better consistency through centralized management
Data sharing	Limited sharing btw applications	Multiple user and applications can share data.
Security	Security is managed seprately by apps	DBMS provides centralised security mech
Data Independence	Applications depend heavily on file structure	Provides logical & physical data independence

Data Integrity	Difficult to enforce common rules	Contains maintain data integrity.
Concurrent Access	Difficult to manage simultaneous access	DBMS provides concurrency control
Back up & recovery	Usually handled seprately	DBMS provides backup & recovery facility.
Data Retrieval	Requires app-specific file handling → suitable for small applications (simple)	SQL provides efficient data retrieval → suitable for large applications that need efficient data sharing, security, consistency etc

Limitations of File Processing Approach:

→ Data Redundancy : same info may be stored in multiple files

→ Data Inconsistency : Updating information in one file may not update other files.

→ Data Isolation : Data is scattered across different files & formats

→ Security Problems: Maintaining security across multiple applications is difficult.

→ Concurrent Access Problems: Simultaneous modifications may lead to incorrect data.

→ Backup and Recovery : Recovering data after system failure can be difficult.

Advantages of Database Approach:

→ Reduce unnecessary data redundancy.

→ Improves data consistency.

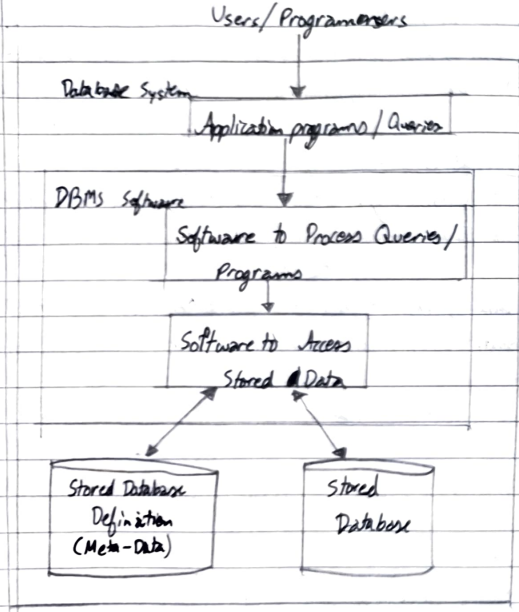
→ Provides data security and access control.

→ Maintains data integrity using constraints

→ Supports concurrent access

→ Allows efficient data retrieval using SQL

Q-4. Explain a database system environment with a block diagram.



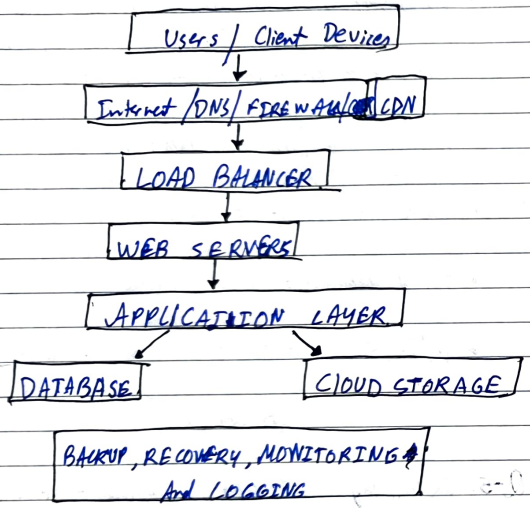
- User/programmers: people who interact with the database to retrieve, insert, delete or modify data.
- Application programs: programs or queries used to send requests to the DBMS.
- DBMS Software: manages the database & process requests from user & application programs.
- Software to process queries/programs: interprets and processes queries or application requests.
- Software to access stored data: accesses the required database records and metadata to perform operations.
- Stored database definition (metadata): Contain info describing the database structure, data type and constraints.
- Stored database: contains the actual data, such as student details, course information & examination records.
- Database Systems: The complete environment containing the DBMS software, stored database, metadata & interfaces used by user/programmers.

Q-5. Sketch the typical development architecture of a cloud application and explain the purpose & functioning of each section.

→ A cloud application is an application hosted on cloud infrastructure and accessed by users through a network.

→ cloud application architecture consists of multiple layers that work together to provide application services, data storage, security and communication.

Block Diagram



Components

1. Client Layer :
   - ↳ It consists of devices and interfaces through which users access the application
   - ↳ sends request to cloud application and displays the response received.

2. → Internet and DNS Layer :
   - ↳ the internet provides connectivity between users and cloud infrastructure.
   - ↳ DNS (Domain Name System) converts domain names into IP addresses so that user devices can locate the application server.

→ CDN (Content Delivery Network);
- ↳ it is a distributed network of servers that delivers frequently accessed static contents, such as images, videos, CSS, and Java Script files, from locations closer to users.
- ↳ reduces loading time and network traffic on the main server.

→ Firewall and Security Layer:
- ↳ This layer protects the application from unauthorized access and malicious traffic. It includes firewalls, authentication, authorization, encryption and access control mechanisms.

3. Load Balancer
   - ↳ distributes incoming user requests across multiple available servers.
   - ↳ improves performance, scalability & availability by preventing a single server from becoming overloaded.

4. Web Server Layer:
↳ handles HTTP/HTTPS requests, serves static content and forwards application request to the application layer. e.g - Apache, Nginx.

5. Application Layer:
↳ Contains business logic of the application. It processes requests, performs calculations, validates data and communicates with databases.

6. Database Layer:
↳ Stores and manages structured or semi-structured application data.
↳ It supports data insertion, retrieval, modification & deletion. while maintaining data integrity. ex - MySQL, MongoDB etc.

→ Cloud Storage:
↳ used to store files such as images, documents, videos and backup. Example - Amazon S3, Azure Blob storage.

7 → Backup and Recovery.
↳ protects application data against accidental deletion, system failure and data corruption. Through backups, snapshots, & recovery mechanisms.

Monitoring & Logging: → help identify and resolve problems
↳ track application performance, errors, server availability, resource usage & database performance.

Q-6 Illustrate the important principles to be addressed from the Design consideration of cloud applications.

→ Cloud Application must be designed carefully to ensure they are secure, scalable, reliable, efficient and cost-effective.

The following principles should be considered during cloud application

1. Scalability
↳ ability of application to handle increase/decrease of workload by adjusting computing resources.
types → Vertical scaling
↳ increase resources of current/existing server, like CPU, RAM & storage.
→ Horizontal scaling
↳ Adding more servers or application instances to distribute the workload.

2. Availability
↳ ability of application to remain accessible & operational whenever required.

3. Reliability & Fault Tolerance
↳ ability to perform its functions correctly & consistently over time.
↳ ability to continue operating even when one or more component fail.

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4. Security
   ↳ ensures that application resources and user data are protected from unauthorized access & attacks.

5. Performance & Efficiency
   ↳ Performance refers to how quickly and effectively an application processes request.

6. Cost Optimization
   ↳ Cloud applications should be designed to minimise unnecessary resource consumption and operational expenses.

7. Maintainability
   ↳ Simplifies application modification and debugging.

8. Loose Coupling
   ↳ Reduces dependencies b/w components.

9. Data Management
   ↳ Maintain data integrity.

10. Monitoring & Logging
    ↳ Track application health & error.